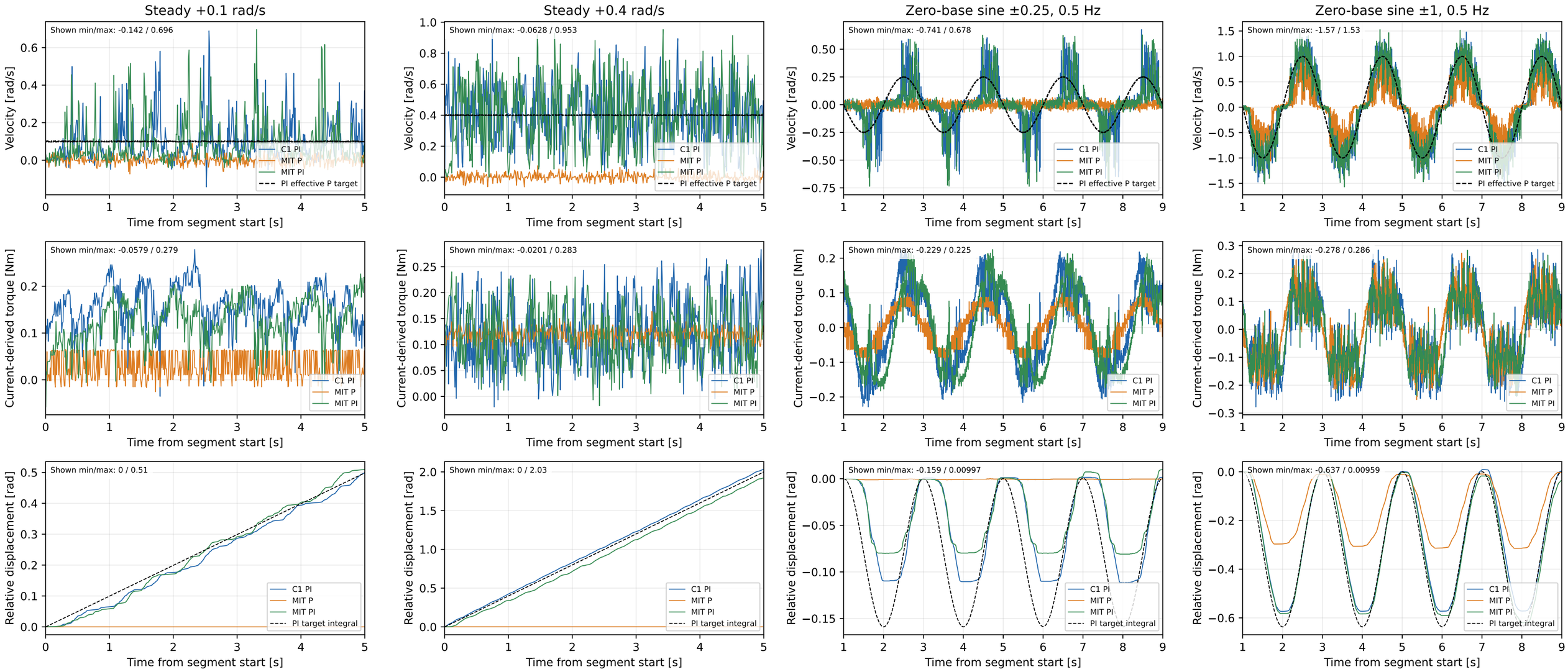

Matched targets, repeat 3: native speed, current-derived torque and encoder travel
Steady +0.1 rad/s
Steady +0.4 rad/s
Zero-base sine ±0.25, 0.5 Hz
Zero-base sine ±1, 0.5 Hz
Shown min/max: -0.142 / 0.696
Shown min/max: -0.0628 / 0.953
Shown min/max: -0.741 / 0.678
Shown min/max: -1.57 / 1.53
Velocity [rad/s]
Shown min/max: -0.0579 / 0.279
Shown min/max: -0.0201 / 0.283
Shown min/max: -0.229 / 0.225
Shown min/max: -0.278 / 0.286
Current-derived torque [Nm]
Shown min/max: 0 / 0.51
Shown min/max: 0 / 2.03
Shown min/max: -0.159 / 0.00997
Shown min/max: -0.637 / 0.00959
Relative displacement [rad]
Time from segment start [s]
C1 PI
MIT P
MIT PI
PI effective P target
PI target integral
Separate runs; no smoothing or fitted time/bias shift. Each A3 subtracts its own first shown sample. Same intended targets; the plotted reference is PI's codec-corrected P target.